$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+\,\alpha\beta} & \mathcal{K}^{\#2}_{1^+\,\alpha\beta} & \mathcal{K}^{\#1}_{1^-\,\alpha} & \mathcal{K}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{Y_1 k^2}{2} & -\frac{Y_1 k^2}{2\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{Y_1 k^2}{2\sqrt{2}} & \frac{Y_1 k^2}{4}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+\,\alpha\beta} & \mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{3 k^2 \, {}^{(4)}\kappa_{15}}{2} & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{3 k^2 \, {}^{(4)}\kappa_{15}}{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+\,\alpha\beta} & \mathcal{J}^{\#2}_{1^+\,\alpha\beta} & \mathcal{J}^{\#1}_{1^-\,\alpha} & \mathcal{J}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{2}{3\,\mathrm{Det}(1^-)} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} & \frac{1}{3\,\mathrm{Det}(1^-)}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+\,\alpha\beta} & \mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{2}{3 k^2 \, {}^{(4)}\kappa_{15}} & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{2}{3 k^2 \, {}^{(4)}\kappa_{15}}
\end{array}
$$

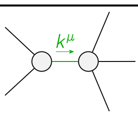
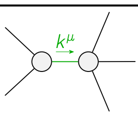
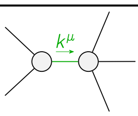
Abbreviations used in matrices

$$Y_1 == 2\,{}^{(4)}\kappa_1 + {}^{(4)}\kappa_{15} \;\&\&\; \mathrm{Det}(1^-) == \frac{3}{4} k^2 \left(2\,{}^{(4)}\kappa_1 + {}^{(4)}\kappa_{15}\right)$$

Lagrangian

$$
\begin{aligned}
&{}^{(4)}\kappa_1\, \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} - {}^{(4)}\kappa_1\, \partial_\chi \mathcal{K}^{\delta}{}_{\beta\delta}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_{15}\, \partial_\chi \mathcal{K}^{\delta}{}_{\beta\delta}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \\
&\frac{1}{2}\,{}^{(4)}\kappa_{15}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} + 2\,{}^{(4)}\kappa_{15}\, \partial_\alpha \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\beta\chi}{}^{\delta} - \frac{1}{2}\,{}^{(4)}\kappa_{15}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - \\
&3\,{}^{(4)}\kappa_{15}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\, \partial_\delta \mathcal{K}_{\chi\beta}{}^{\delta} - {}^{(4)}\kappa_{15}\, \partial_\alpha \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + {}^{(4)}\kappa_{15}\, \partial_\delta \mathcal{K}_{\alpha\beta\chi}\, \partial^\delta \mathcal{K}^{\alpha\beta\chi} + {}^{(4)}\kappa_{15}\, \partial_\delta \mathcal{K}_{\beta\alpha\chi}\, \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}
$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{0^-} == 0$	1	$\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta \mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#1\,\alpha}_{1^-} + 2\,\mathcal{J}^{\#2\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#2\,\alpha\beta}_{1^+} == 0$	3	$\partial_\delta\partial_\chi\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\delta\partial_\chi\partial^\beta \mathcal{J}^{\chi\alpha\delta}$
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == 0$	3	$\partial_\delta\partial_\chi\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial_\chi\partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi \mathcal{J}^{\alpha\beta\chi}$
Total # constraint(s):	11	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{2\,{}^{(4)}\kappa_1 + {}^{(4)}\kappa_{15}}$
	2	0	$-\frac{1}{{}^{(4)}\kappa_{15}}$
	2	0	$\frac{1}{{}^{(4)}\kappa_{15}}$
Resolved unitarity condition(s):		(Demonstrably impossible)	